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QUANTITY DISCOUNTS: MANAGERIAL ISSUES AND RESEARCH OPPORTUNITIES

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Marketing managers frequently administer pricing policies involving substantial quantity discounts. However, our understanding of the precise role of these discounts, and their design, is quite limited. This paper reviews and integrates the economics and marketing literature and presents a detailed analysis of the policies of four firms in order to: (i) specify the variety of motivations for quantity discounts, (ii) provide guidelines to managers in designing quantity discount schedules, and (iii) set out the most fruitful directions for future research.

(Pricing; Discounts; Price Discrimination)

1. Introduction

A pricing policy involving substantial quantity discounts is a common phenomenon. As consumers, for example, we have all seen advertisements like the following:

- “Twenty percent off all cleaning bills over \$9.00” at the local laundry.
- “Buy the series and get one show free” from the theater group.
- “Join the family shoe plan. Buy 10 pairs and get one pair free” at the shoe store.
- “Buy a case of assorted bottles and get 20% off” at the wine store.

Nason and Della Bitta's (1983) survey shows consumers expect quantity discounts in the form of lower unit prices for larger package sizes of a given brand. Quantity discounts are also common and even expected at other levels in the channel of distribution and in industrial markets. For example, GTE's winter 1984 advertising theme to businesses for its Sprint long-distance telephone service was: “What, you're not getting volume discounts?”

Still, there is much confusion about the principles to follow in designing an implementable and profitable discount schedule. When asked about quantity discount policy, one marketing manager replied: “Well, I'm not sure I know why we do it that way except that that's the way it was printed in the pricing book when I arrived.” Another common reply was: “That's the way *X* (a competitor) does it and we have to be competitive.” Finally, a management consultant specializing in the grocery business summarized practice in that industry by saying, “I've never seen one (a discount schedule) that makes sense.”

Given this situation our purpose here is dual:

1. To improve managerial decision making in the development of pricing schedules, by providing guidelines from the economics and marketing literature on quantity discounts.

2. To stimulate normative research work by identifying the key concerns of management, and by providing a detailed description of quantity discount schedules employed in four firms. In each case, we determine the motivation for the discount schedule, and we analyze the four key characteristics of a schedule:

- a. Form, i.e., whether the discount applies to all units or only units incremental to certain amounts.
 - b. Complexity, i.e., the number of price break points in the schedule.
 - c. Depth, i.e., the size of the discount at each break point.
 - d. Qualifying units, i.e., the aggregation permitted in determining the discount.
- Schedules may be based on quantity of a single item ordered at one time, aggregated across items or across time.

The paper is organized as follows: §2 presents terminology and establishes the relationship among different types of price schedules. §3 reviews the literature from economics and marketing, highlighting motivations for quantity discounts and key results. §4 discusses the practices of four firms and relates practice to the literature discussion of §3. §5 concludes the paper with discussion of key design and implementation issues.

2. Terminology

Since different names are often used to refer to equivalent price schedules, it is important to understand the relationship among alternative schedules. The first differentiating characteristic of a price schedule is whether it is a *uniform* or a *nonlinear* (or nonuniform) schedule. Let $R(q)$ be the charge for an amount q . A uniform price schedule is one in which there is a linear, through the origin, relationship between q and $R(q)$, i.e., the average price paid is independent of q . See Figure 1(a). Any price schedule not satisfying this condition is nonlinear. Common nonlinear schedules are:

(i) *A two-part tariff* (Figure 1(b)): essentially, the customer pays for the right to purchase the good since a fixed charge $F > 0$ is imposed if any good is purchased and a uniform price p is then paid for all units,

$$R(q) = \begin{cases} F + pq, & q > 0, \\ 0, & q = 0, \end{cases} \quad \text{where } F > 0. \quad (1)$$

Note that although the customer pays the same *marginal* unit price for all quantities, the *average* price paid is a monotonically decreasing function of quantity purchased.

(ii) *A two-block tariff* (Figure 1(c)): a per-unit price p_1 is charged for any units, up to quantity x , then per-unit price changes to p_2 for all units greater than x ,

$$R(q) = \begin{cases} p_1 q, & 0 \leq q \leq x, \\ p_1 x + p_2(q - x) & \text{for } q > x. \end{cases} \quad (2)$$

Note that unlike the constant marginal unit price for two-part tariffs, the marginal unit price changes at quantity x for a two block tariff.

The two-block tariff with $p_2 < p_1$ is a quantity discount. Specifically, a schedule such as (2) would be an incremental quantity discount with one price break at x . “ N -part” and “ n -block” tariffs with $n > 2$ are defined similarly.

(iii) *All-units quantity discount* (Figure 1(d)) yields an $R(q)$ function with discontinuities. In this schedule, if a certain quantity level is exceeded, the price applies to all units, not just those incremental to the break point, viz.

$$R(q) = \begin{cases} p_1 q & \text{if } 0 \leq q < x, \\ p_2 q & \text{if } q \geq x, \end{cases} \quad \text{where } p_1 > p_2. \quad (3)$$

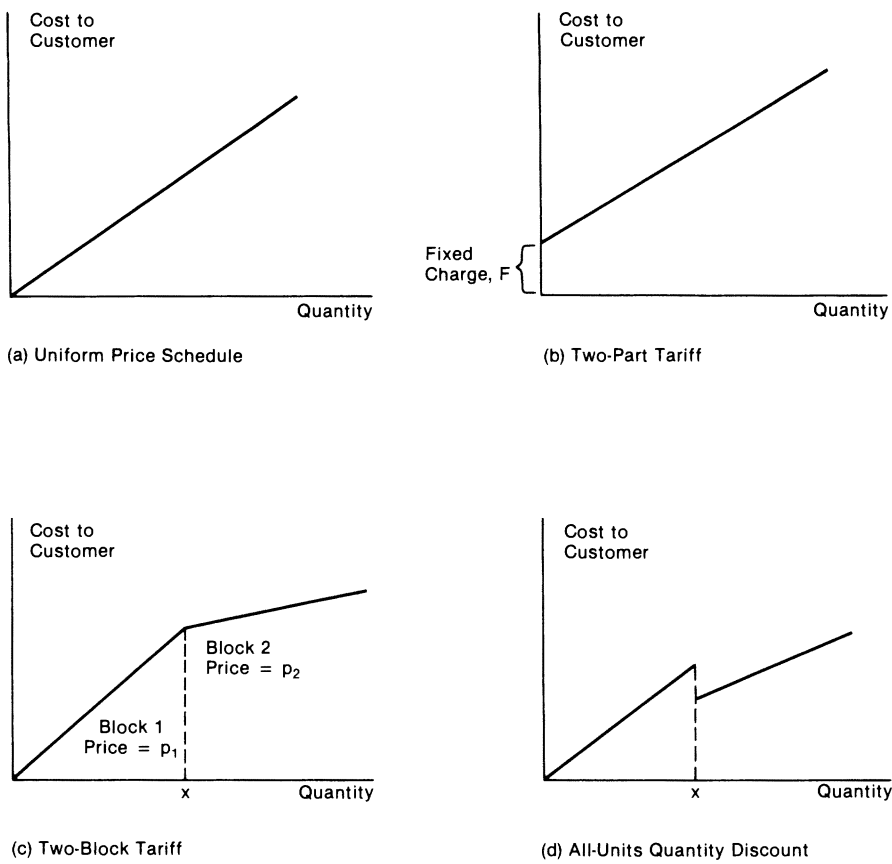


FIGURE 1. Price Schedule Illustrations.

Relationships Among Tariffs

The following relationships hold:

(i) As is evident from Figure 1(a), a uniform price schedule is a special case of both a two-part tariff ($F = 0$ in (1)) and a two-block tariff ($p_1 = p_2$ in (2)).

(ii) Under general conditions, two-block tariffs are better than two-part, i.e. the optimal two-block tariff yields profits greater than or equal to the optimal two-part tariff. If there exists a finite price $\bar{p}$ such that no customer values a unit of the good at more than $\bar{p}$, then the three parameters of two-block schedule (p_1, p_2, x) can be written as a function of the two parameters of the two-part tariff (F, p) and $\bar{p}$ to yield the same quantity purchased and revenue from each customer. Specifically, the two-block tariff defined by:

$$p_1 = \bar{p}, \quad p_2 = p, \quad x = F/(p_1 - p_2)$$

is equivalent to the two-part tariff (F, p). This result is due to Leland and Meyer (1976, p. 455).

(iii) While (ii) states that any two-part tariff can be written as an equivalent two-block tariff, the converse is not true in general. Leland and Meyer (1976, Table 1, p. 459) give examples for heterogeneous customers where the optimal two-block tariff yields strictly greater profits than the optimal two-part. The advantage for two-block tariffs derives from the fact that under such a schedule, different customers can pay different marginal prices. In a two-part tariff, all customers pay the same marginal price albeit different average prices. However, a set of two-part tariffs with each customer selecting one or the other can be designed to be equivalent to a two-block schedule.

(iv) There is a situation in which a two-block tariff can be written as an equivalent two-part tariff. This condition is that *all* customers are paying the same marginal price (i.e., purchasing quantities in the same block). This is evident from the graph of a two-block tariff in Figure 1(c). If all customers are purchasing quantities less than x , then the marginal price p_1 applies for all. Customers would demand the same quantities and pay the same amount under the uniform pricing policy $p = p_1$. If all customers are purchasing quantities greater than or equal to x , the two-part tariff $F = (p_1 - p_2)x$, $p = p_2$ yields identical quantities demanded and revenues. This argument applies to a n -block tariff with $n > 2$ if all customers are purchasing in the same block as well. Moorthy (1987) applies this idea in his commentary on Jeuland and Shugan's (1983) channel coordination work to show that their quantity discount schedule can be replaced by an equivalent two-part tariff.

An understanding of the relationship among schedules is important because schedules differ in their implementation costs. For example, a two-part tariff does not require the firm to monitor consumption over a time period to determine marginal prices, whereas the two-block tariff does.

3. Literature Review

Organization of the Review

Buchanan (1953) presents three principal motivations for quantity discounts. They may be a mechanism for:

- (i) achieving perfect price discrimination against a single customer or a set of homogeneous customers;
- (ii) achieving partial price discrimination against a set of heterogeneous customers;
- (iii) influencing the buyer's ordering pattern, i.e., increasing the quantity per order given a total quantity demanded. This shifts some of the inventory holding cost from the seller to the buyer and can increase the efficiency of the buyer-seller system.

Since the work focusing on the first motivation (Buchanan 1953 and Gabor 1955), economists have concentrated on the second. The less extensive work in marketing has focused on the third motivation. Table 1 gives the motivations and principal related papers.

In Cases 1 and 2, the firm faces a set of downward sloping demand curves and maximizes profit by: (i) having each customer buy the quantity at which his marginal value equals the firm's marginal cost (this maximizes welfare) and (ii) having the total charge for the quantity equal the total value to the customer (this yields zero consumer surplus). A quantity discount dominates a uniform pricing policy in approaching these profit maximization conditions. For Case 3, the first five papers consider quantity discounts as a means to coordinate the ordering and inventorying pattern of members of the channel of distribution. Like Cases 1 and 2, the objective is to increase the total surplus, i.e., benefits-costs, of the system composed of the buyer and the seller. The distinguishing characteristic of Case 3 is that the discount is intended to influence the quantity per order, but not the total quantity demanded over the long run. Jeuland and Shugan (1983), McGuire and Staelin (1986), and Moorthy (1987) consider a somewhat different system efficiency motivation detailed below. We now review each case.

Case 1: Homogeneous Customers

Buchanan (1953) and Gabor (1955) consider the case of an individual customer or a homogeneous set of customers. Figure 2 shows the two negative characteristics of the optimal uniform price P_u and associated quantity demanded Q_u . First, the customer's valuation of Q_u units exceeds the price he must pay, yielding the positive consumer surplus shown. Second, the quantity demanded, Q_u , is less than Q_w . At Q_w , the cus-

TABLE 1
Principal Papers Classified By Motivation Considered

	Case 1	Case 2	Case 3
Motivation	perfect price discrimination	partial price discrimination	channel efficiency
Setting	homogeneous buyers	heterogeneous buyers	general
Principal Papers	Buchanan (1953) Gabor (1955) Moorthy (1985)	Oi (1971) Leland and Meyer (1976) Faulhaber and Panzar (1977) Murphy (1977) Spence (1977) Goldman, Leland, Sibley (1977) Oren, Smith and Wilson (1982), (1983)	Crowther (1964) Dolan (1978) Dolan (1980) Lal and Staelin (1984) Monahan (1984) Jeuland and Shugan (1983) McGuire and Staelin (1986) Moorthy (1987)

tomer’s marginal value of the good equals the firm’s marginal cost. At quantities less than Q_w , such as Q_u , the value the customer places on the good is greater than the marginal cost. Q_w is the “welfare maximizing” quantity and P_u results in $Q_u < Q_w$ and the welfare loss shown.

Buchanan’s solution is a multiblock price schedule. He shows that setting price equal to the customer’s marginal evaluation at each quantity achieves welfare maximization (i.e., quantity purchased Q_w) and zero consumer surplus. Gabor (1955) shows that the complexity of a multiblock schedule is not required to achieve this result. He shows that a two-block tariff (i.e., an incremental quantity discount with one price break) is sufficient.¹ As noted above in the Relationships Among Tariffs section, Moorthy (1987) shows that a single two-part tariff can achieve the same result as any multiblock tariff given homogeneous customers. In summary, two points are key from Case 1 literature: (i) a quantity discount schedule or two-part tariff can offer significant benefit over a uniform price schedule, (ii) achieving the welfare maximizing level of consumption with zero consumer surplus does not require the complexity of a multiblock schedule as Buchanan suggested. For the homogeneous customer case, either a two-block or a two-part schedule can achieve the desired result.

Case 2: *Heterogeneous Customers*

A second motivation for quantity discounts is partial price discrimination among heterogeneous customers. The underlying motivation for this is that “the demand schedule of small buyers is more inelastic over the relevant price range than that of large buyers” (Buchanan 1953). While Buchanan provides the motivation and states the necessary condition, Oi (1971) solves the problem of price schedule design for this case.

Oi specifically considers two-part tariffs. His work stimulated considerable research on other tariff forms as well. We begin with a review of Oi’s analysis. Figure 3 shows demand curves for a “large” and “small” buyer and assumed constant marginal cost for the firm. If the firm could identify and have a different price schedule for each buyer, the profit maximizing result of (i) the small and large buyer purchasing the welfare

¹ Essentially, this is achieved by setting the price in the second block p_2 at marginal cost and setting price in the first block, p_1 , at a high price to extract all the consumer surplus.

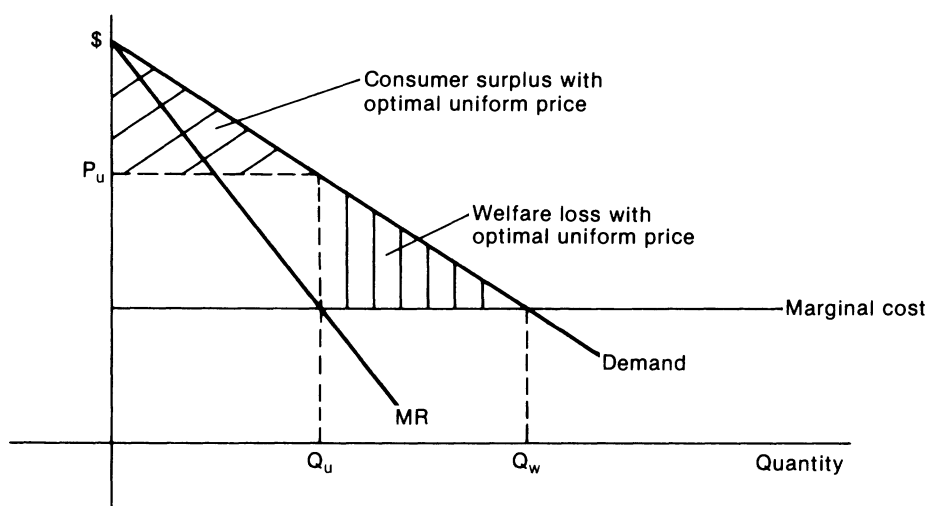


FIGURE 2. Individual Demand Curve.

maximizing quantities Q_{sw} and Q_{tw} , respectively, and (ii) each having zero consumer surplus, could be achieved. Two-part tariffs with price equal to marginal cost and fixed payments equal to the areas under the demand curves as shown in Figure 3 achieve this result.² With the constraint of a single marginal price schedule uniformly applied to all customers, this solution is infeasible since the large buyer's fixed payment is greater than the small buyer's. One feasible solution for the single marginal price schedule problem is to keep price equal to marginal cost, but have the fixed charge for the small customer (the $\square$ area in Figure 3) apply to both customers. However, Oi shows that for customer demand curves which do not cross, as shown in Figure 3, the profit maximizing solution is to raise price above marginal cost and decrease the fixed charge to keep the small buyer's surplus nonnegative.³ Properties of this optimal solution, as seen from Figure 4 are:

- (i) each buyer purchases less than his welfare maximizing quantity because price P^* is above marginal cost;
- (ii) the large buyer has positive consumer surplus.

These limitations of a two-part tariff stimulated significant research into alternative, more complex price schedules. This work considered the total cost function, $R(q)$, as both a piecewise linear function and as a continuous, nonlinear function. We now review the work on discrete and continuous schedules in turn.

Discrete Price Schedules

Major papers on discrete schedules are Leland and Meyer (1976), Murphy (1977), and Faulhaber and Panzar (1977). As discussed above, Leland and Meyer's key result is a comparison of two-part and two-block schedules, viz. (i) any two-part tariff can be replaced by a two-block tariff yielding the same quantity purchased and revenue received from each customer and (ii) a two-block can dominate a two-part tariff if customers are heterogeneous. In short, Leland and Meyer show that, implementation

² The assumptions here and generally throughout the nonlinear tariff work are: (i) zero income elasticity, i.e., the size of the fixed fee does not affect the demand for the good and (ii) infeasibility of resale. That is, one buyer cannot pay the fixed fee then sell to others as well.

³ In the case where the small buyer is "much smaller" than the large buyer, the optimal solution may be to set prices to price discriminate perfectly against the large buyer and sell nothing to the small buyer. We assume that this is not the case here, although Oi does analyze this case.

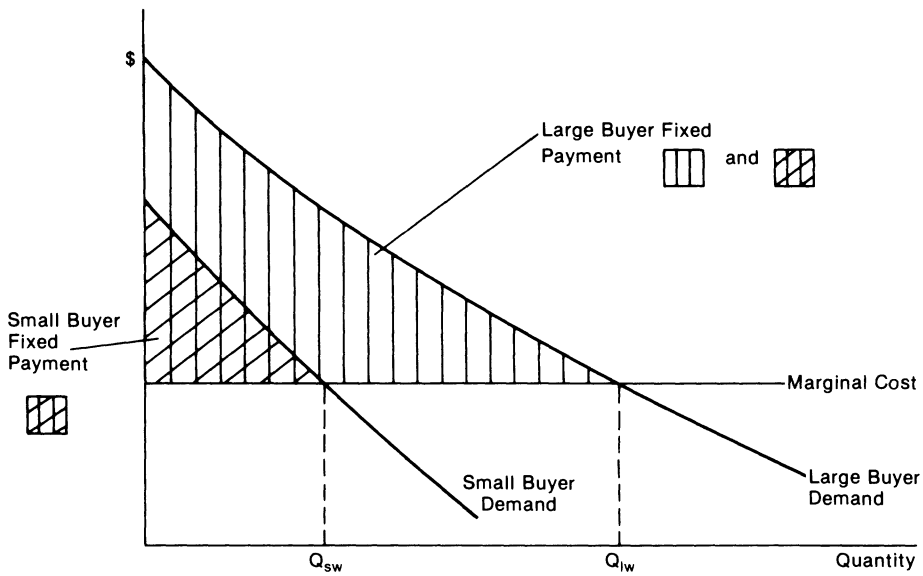


FIGURE 3. Heterogeneous Customer Demand.

costs aside, a schedule with two marginal prices can do better than a schedule with a fixed fee and one marginal price.

If two marginal prices are better than one, are three marginal prices better than two? Murphy (1977) answers affirmatively, given heterogeneous customers. If customer i 's demand curve is given by $Q_i(p) = A_i - p$, where A_i is some continuous distribution, he shows:

- (i) profit is *strictly* increasing in the number of blocks in a schedule, but at a decreasing rate;
- (ii) the incremental value of an additional block is increasing in the heterogeneity of customers;

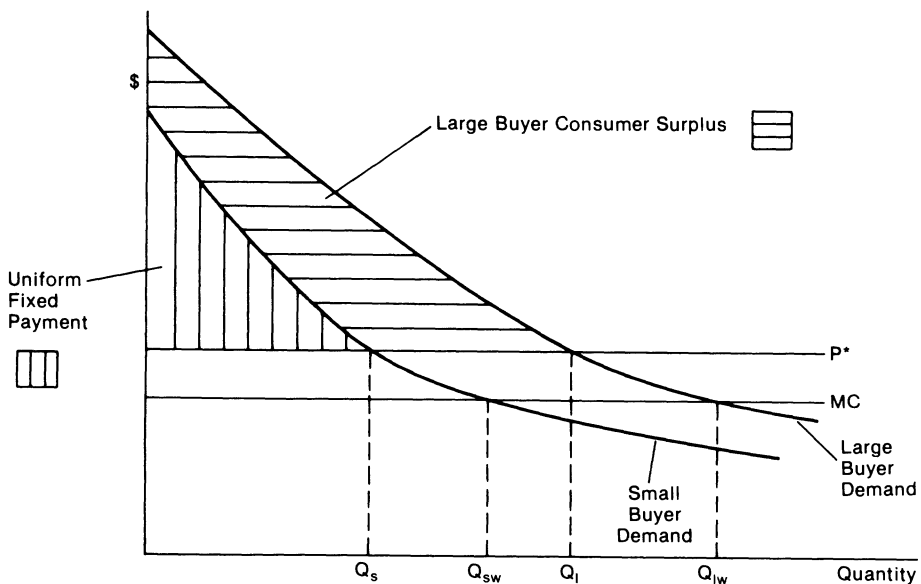


FIGURE 4. Optimal Uniform Price Schedule for Heterogeneous Customers.

(iii) the magnitude of price decline in an optimal schedule is constant across all blocks, i.e., $p_i - p_{i+1} = C \forall i$.

(iv) the optimal rate of price decline increases in the heterogeneity of customers.

While Murphy establishes the benefit of multiblock schedules, such schedules can involve significant implementation cost, e.g., in communicating the schedule, monitoring each customer's quantity purchased, and adjusting prices each time a new block is reached. These implementation considerations make the Faulhaber and Panzar (1977) results important. They show that customer self-selection from a set of n two-part tariffs is equivalent to the imposition of an n -block schedule. The benefit of the n two-part tariffs is that a given customer pays a fixed fee and then the same marginal price for all units purchased. Hence, the implementation costs are significantly less. We often see self-selection from two-part tariffs as a "nonmember" schedule $(0, p_1)$ and a member schedule (F, p_2) where members pay F once to receive the lower p_2 unit price.

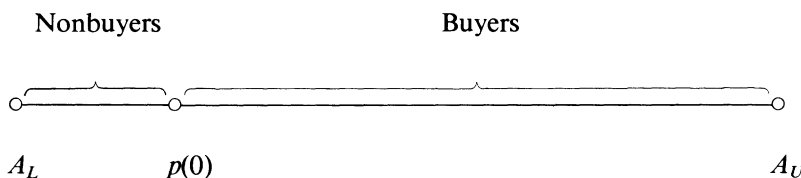
Continuous Price Schedules

Taking $R(q)$ to be discrete or continuous represents a trade-off between implementation concerns and power of analytical results obtainable. By specifying $R(q)$ as a continuous function, Spence (1977), Goldman, Leland and Sibley (1977), and Oren, Smith and Wilson (1982, 1983) achieve powerful insights into the qualitative properties of the optimal price schedule. In addition, Oren, Smith and Wilson (1983) present results for a competitive situation.

The primary result of this work is that under the assumption of no income effects, determining the price schedule, $R(q)$, can be considered as a series of *independent* pricing decisions at each quantity level. For ease of exposition, consider a demand situation similar to Murphy (1977) viz. the demand curve for customer i is

$$Q_i(p) = A_i - p$$

where A_i is uniformly distributed on $[A_L, A_U]$. Figure 5 shows the form of this demand heterogeneity by plotting the demand curve for several buyers. Each buyer i on $[A_L, A_U]$ values the q th unit of the good at $A_i - q$ up to $q = A_i$, thereafter the marginal value is zero. Thus, the values customers place on the first unit of the good ($q = 0$) are uniformly distributed on $[A_L, A_U]$. The proportion of the market buying the first unit is determined by the price. Any price, $p(0)$, greater than the lower bound on values, A_L , separates the market into buyers and nonbuyers as shown:



Thus, the percentage of the market buying if $p(0)$ is charged is: $(A_U - p(0))/(A_U - A_L)$. The profit maximizing firm selects $p(0)$ to maximize this percentage times the associated unit margin, i.e., assuming constant marginal cost c , the firm maximizes:

$$\frac{A_U - p(0)}{A_U - A_L} (p(0) - c).$$

At any quantity q , the marginal price $p(q)$ is determined in a similar fashion. The important point is that, given the assumption of no income effects, the optimal $p(q)$ is independent of conditions at other quantity levels. In general, the optimal price at quantity level q is given by:

$$p^*(q) = \frac{c}{[1 - 1/e(q)]} \quad (4)$$

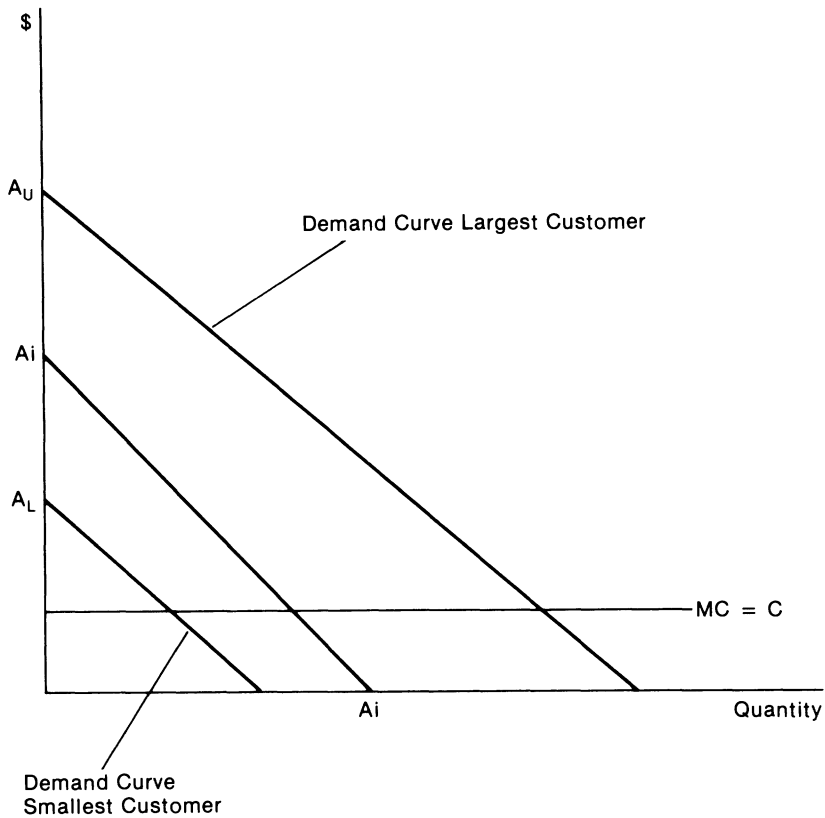


FIGURE 5. Continuous Price Schedule Determination for Specified Demand Heterogeneity.

where $e(q)$ is the elasticity of the market demand curve at quantity q . In short, the classical monopoly profit maximizing condition must hold simultaneously at each possible consumption level. This condition yields insight into Murphy's numerical results for discrete schedules which show the rate of price decline increasing with the heterogeneity of the population. This is equivalent to the elasticity, $e(q)$, changing over quantity levels. Oren, Smith and Wilson (1982) extend the results of the earlier papers to include cases of demand externalities, i.e., where the value a customer places on the product depends on the number of other consumers purchasing.

More important, Oren, Smith and Wilson (1983) derive an optimality condition for a competitive setting. They show that if quantity discounts are optimal in monopoly, they are optimal in competition among a finite number of identical firms as well. In the simple Cournot equilibrium with n identical firms selling at a uniform price, the optimal price for quantity q is:

$$p^*(q) = \frac{C}{\left[1 - \frac{1}{ne(q)}\right]} \quad \text{where } n = \text{number of competitors.} \tag{5}$$

Thus, if the market conditions (captured in $e(q)$) are such that quantity discounts are optimal in the monopoly case ($n = 1$), they are optimal for any finite number of competitors. As the number of competitors approaches infinity, the situation approaches perfect competition and the associated well-known uniform price at marginal cost result.

Summary of Cases 1 and 2

The research reviewed to this point does not seek to provide implementable pricing algorithms. Rather, its aim is to provide understanding of the role of quantity discounts and insight into the considerations influencing their design. The understanding they provide is as follows:

- (i) under general demand conditions, the profit maximizing price schedule is non-linear;
- (ii) customer heterogeneity is the key in determining the benefits of
 - (a) nonlinear tariffs as compared to a uniform price schedule and
 - (b) nonlinear tariff of varying degrees of complexity.

The examples below in §4 will both demonstrate the contribution of this literature to practice and establish its limit.

Case 3: Channel Efficiency

In contrast to the economics literature's price discrimination motivation, the marketing literature presents quantity discounts as a coordinating mechanism in the channels of distribution. The major type of coordination considered is purchase timing. Second, Jeuland and Shugan (1983) show quantity discounts are a mechanism for more generally coordinating marketing policies of a manufacturer and a retailer.

Crowther (1964) establishes the purchase timing motivation. For this motivation, an individual's demand curve is assumed perfectly inelastic at demand level D per time period. In purchasing D units, the buyer selects the quantity per order Q trading off the cost of making D/Q orders and the cost of carrying on average inventory of $Q/2$, viz.

$$\text{Minimize } \frac{D}{Q}A + \frac{Q}{2}I \quad \text{where} \quad (6)$$

Q = quantity per order,

A = buyer's cost to place an order,

I = annual dollar cost of holding an item in inventory.

The "economic order quantity" solving (6) is $\sqrt{2AD/I}$.

While this quantity is "economic" for the buyer, the solution does not consider the impact of the order size on the seller. Crowther argues that the seller's order processing cost per time period is decreasing in Q and thus quantity discounts can be useful. Dolan (1978) formalizes Crowther's arguments by specifying quantity discounts as a mechanism for inducing the buyer to purchase in quantities minimizing the "system cost," i.e., the sum of the buyer and seller cost. Dolan (1980) determines the optimal discount schedule for the homogeneous customer case, and shows a price schedule with a single break point achieving the desired outcome. This parallels the finding of Gabor (1955) on the sufficiency of single break schedules for the perfect price discrimination motivation with homogeneous customers. Independently of Crowther and Dolan, Monahan (1984) develops a similar analysis of the homogeneous customer case.

Lal and Staelin (1984) advance understanding of the channel efficiency motivation by considering heterogeneous buyers and an ordering cost which is a continuous function of Q . While they do not specify the optimal discount schedule as a function of buyer and seller parameters, they present a numerical search algorithm to find the optimal price schedule.

Related to this coordination of ordering/inventory behavior is the marketing program coordination motivation proposed by Jeuland and Shugan (1983). Consider a manufacturer setting a uniform price to a retailer who in turn sets a uniform market price. Jeuland and Shugan show that individual profit maximizing behavior by each member of the channel does not lead to the joint profit maximizing market price. They

show that a quantity discount from the manufacturer is a way to achieve the channel coordination necessary to arrive at the joint profit maximizing market price. McGuire and Staelin (1986) and Moorthy (1987) attempt to show how the same impact can be achieved in a more easily implementable way. McGuire and Staelin propose an “once-a-period” franchise fee based on retailer profits. Moorthy (1987), in a comment on Jeuland and Shugan, shows the same incentive can be provided via a two-part tariff rather than a quantity discount schedule in the homogeneous retailer case.

Thus, the marketing and economics literatures provide different motivations for quantity discounts. However, the motivations share the objective of maximizing system welfare. An objective of price discrimination is customers purchasing the welfare maximizing quantity, i.e. the quantity equating marginal value and marginal cost. For logistics efficiency, the goal is minimization of total system cost for a given demand level transaction. Structurally, there is a similarity between the price discrimination and Jeuland-Shugan coordination motivations as well. In the perfect price discrimination case, the buyer’s downward sloping demand curve generates quantity discounts as discussed above in Case 1. In the manufacturer/retailer coordination situation of Jeuland and Shugan, the retailer faces a downward sloping demand curve from the market. Hence, the retailer must lower his price to sell more quantity. Implicitly, then, the retailer’s demand curve for the manufacturer’s product is downward sloping. The optimality of manufacturer’s quantity discounts to the retailer then follows from the same argument as made in the perfect price discrimination case.

Summary of Literature Review

The motivations for quantity discounts are made clear in the literature. They are definitely more complex and significant than the superficial notion that discounts are a concession to large buyers. Specifically, the literature establishes four motivations:

1. Perfect price discrimination of a homogeneous buyer group.
2. Partial price discrimination of a heterogeneous buyer group.
3. Logistics system efficiency.
4. Coordination between levels in a channel of distribution.

We now present four examples of actual discount schedules to see how practice and the literature compare.

4. Examples of Alternative Rationales

Here, we analyze the quantity discount practices of four firms to identify key managerial issues in price schedule design and assess the contribution of the literature. The sample of firms is purposive, not random. We present, in turn, an example of logistics system efficiency motivation, two examples of partial price discrimination and, finally, a competitive bidding example which we believe has not yet been studied.

Logistics System Efficiency: Sealed Air Corporation⁴

Sealed Air Corporation sells protective packaging material to distributors for resale to end users. The product has a low dollar value to volume ratio, making transportation and inventory carrying costs an important concern. Sealed Air attempts to differentiate itself as the “national supplier.” In support of this, Sealed Air has eight manufacturing facilities from Massachusetts to California, and an “F.O.B. Destination” condition in its price schedule effectively creating one countrywide price. “F.O.B. Destination” means that title passes to the buyer at the buyer’s location with Sealed Air responsible

⁴ This discussion is based partly on information from Sealed Air Corporation, 9-582-104, Harvard Business School Case Services, written by R. J. Dolan.

for delivery. The resulting concerns about transportation efficiencies afforded by large individual shipments are reflected in Sealed Air’s price schedule which is as follows:

Quantity Purchased in Square Feet	Price Per Thousand Square Feet
1,000– 5,000	\$50.00
5,000–10,000	45.40
10,000–30,000	40.90
30,000–50,000	38.10
50,000–truckload	37.05
truckload (approximately 200,000 sq. ft.)	33.50

where “quantity purchased” is defined as the “total square footage, of a single order combining all grades, *ordered for shipment at one time to a single destination.*” [Author’s emphasis]

The quantity discounts induce the majority of customers to buy in truckload quantities. This has the dual effect of reducing Sealed Air’s transportation cost and increasing the average inventory of the distributors. If the policy were “F.O.B. Factory”, then the transportation cost burden would fall on the buyer and there would be less incentive for Sealed Air to offer quantity discounts. However, given the desire to quote a nationwide delivered price, Sealed Air quotes “F.O.B. Destination.” The quantity discounts are a mechanism of sharing the transportation cost savings and shifting the inventorying responsibility for a bulky item down the distribution channel.

Partial Price Discrimination: Two Examples

EXAMPLE 1: IBM Computer Memory.⁵ In June 1980, IBM introduced a new generation of computer memory with the 3380 disk drive. This technological advance meant that IBM essentially would have a monopoly until the plug-compatible manufacturers could develop substitutes. Expecting competition in early 1983, IBM changed its pricing policy in December 1982:

- (1) It cut 3380 prices approximately 15% (e.g., the Model AAY price was cut from \$116,050 to \$98,640).
- (2) It offered “customers who promised to buy 10 or more disk drives over the next two years” additional discounts of 6–12% (*Business Week*, 1983). Specifically, the discount schedule was: 6% for quantities of 10–29, 9% for 30–49, and 12% over 49, where quantity is based on “promises to buy over the next two years.”

Why the quantity discounts? As discussed in Case 2 above, the partial price discrimination literature shows quantity discounts are warranted if the demand schedule of the small buyer is more elastic than that of the large buyer, i.e. if there is a negative correlation between willingness to pay (price sensitivity) and size of buyer (measured in volume purchased). Does this apply to the IBM situation? The simple model which follows shows that it does. We also specify the source of the required correlation.

Consider an individual customer using an incumbent firm’s product but still having *N* units to be purchased when the entrant’s product becomes available. Assume that demand is price inelastic, meaning *N* units will be purchased regardless of the entrant’s price. Let:

- V_e = customer’s valuation of entrant’s product,
- V_f = customer’s valuation of incumbent firm’s product,

⁵ This discussion is partially based on *Electronic News*, “IBM Cuts 3380 Tag 15%; Initiates Volume Discounts,” December 20, 1982, p. 1; *Business Week*, “Memory Makers Lose Ground to IBM,” April 11, 1983, p. 124D.

S = one-time cost to switch from incumbent firm to entrant, e.g., for establishing relationship, checking out equipment, adapting software, etc.

The switching cost, $S > 0$, provides an incentive to stay with the incumbent firm. The buyer remains with the incumbent for the future N units if

NV_f – Cost of N Units from Incumbent
↑
Value of N
Units of
Incumbent's
Product

$\geq NV_e$ – Cost of N Units from Entrant – S
↑
Value of N
Units of
Entrant's
Product

↑
Switching
Cost

(7)

Dividing by N and rewriting (7), no switch is made if:

Average Price of N
Units from Incumbent

$\leq$

Average Price
of N Units from
Entrant

$+ \left\{ (V_f - V_e) + \frac{S}{N} \right\}$

(8)

The bracketed terms on the right-hand side are “the permissible price differential” for the incumbent. (Note: This may be negative since V_f could be less than V_e .) The important aspect of this permissible price differential is that it is decreasing in N , through the S/N term. Thus, if the entrant has a uniform price policy or offers quantity discounts (i.e., a declining average price), the relationship is as shown in Figure 6.

The declining permissible average price per unit provides the impetus for quantity discounts on the part of the incumbent if individuals differ in N , the number of units to be purchased.

EXAMPLE 2: *Los Angeles Times* Retail Display Advertising. In 1981, the *Los Angeles Times* carried more advertising than any other paper in the United States. Its quantity discount schedule involved 32 price breaks and a maximum discount of 40% (for buyers of more than five million lines). Why would the paper offer such significant discounts?

An answer might be cost savings. However, two experts in the newspaper business rule this out: “Indeed, they [the prices for large quantities] are lower than would be warranted by straightforward cost-savings . . .” (Billings 1982, p. 7) and “Almost everyone gives discounts much larger than you can justify on the basis of cost alone” (Brandon 1981, p. 9). The Federal Trade Commission hearings on the price schedule brought forth two other explanations: one specific to the newspaper business and one more general.

First, as argued by Billings in his comments on the FTC consent decree, large firms are valuable to newspapers as advertisers in that consumer interest in their ads increases readership. Advertising by small retailers, however, does not generate similar increased consumer interest in a newspaper. Consequently, there is no incentive for the paper to pursue small retailers other than for the billings they generate. However, the granting of discounts greater than the cost savings is not peculiar to newspapers; e.g., the same phenomenon occurs in the marketing of many industrial products.

The second, more general, explanation is alluded to by Tom Johnson, publisher of the *Los Angeles Times*: “Our present rates have been established in response to our costs and to competition from both newspaper and other media, including broadcast, within our market” (Gloede 1981, p. 11) (Author’s emphasis). Clearly, the feasible options of large advertisers are greater than those of small firms. While the *Los Angeles Times* faces competition from local papers, billboards, and handbills in the case of the small retailer, it faces additional competition from the local TV stations in the case of large firms. The expansion of the set of feasible alternatives with buyer size decreases the large buyer’s willingness to pay for newspaper space, thereby creating the negative

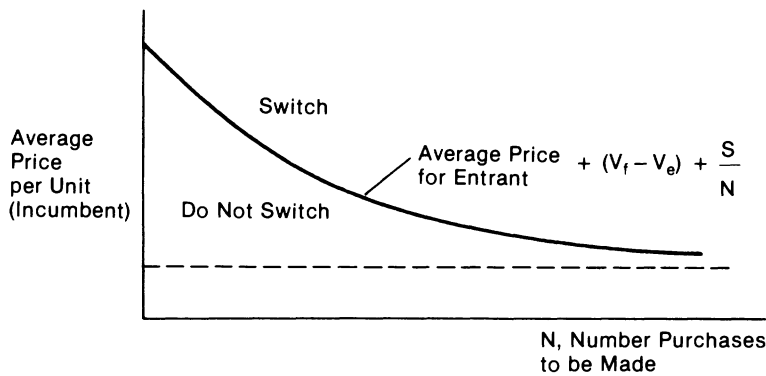


FIGURE 6. Switch/No Switch Zones as Function of Incumbent Price and Number of Purchases to be Made.

correlation between willingness to pay and size which is necessary for the partial price discrimination explanation.⁶

Thus, the IBM and *Los Angeles Times* examples both fit the partial price discrimination motivation. The point of including both here is not merely to suggest some frequency of occurrence of the motivation, but rather to contrast differing sources of the requisite correlation between size and price sensitivity. The correlation in the two cases emanates from different sources. In IBM, customers face identical feasible competitive alternatives, and the amortization of a fixed cost (*S*) generates the required correlation. The *Los Angeles Times* demonstrates that customers facing different feasible competitive alternatives can generate the correlation even when “switching costs” are zero.

Competitive Bidding: Electronic Components Suppliers

In many competitive bidding situations, the buyer asks suppliers to provide prices for various quantities purchased. Suppliers commonly respond with quantity discounts. For example, in procuring its upcoming year’s requirement of 20 million units from semiconductor manufacturers, IBM wished to maintain a competitive, multisourced environment.⁷ Based on previous contracts, the expected division of the business between the two suppliers was 60% to firm 1 and 40% to firm 2. The IBM procurement team requested bids for these levels, and levels plus and minus 10% and 20% of expected. Quotations on price per unit received were:

Percentage of Requirements	Firm 1	Firm 2
80%	\$63.28	—
70%	64.33	—
60% “expected” →	65.48	\$61.74
50%	67.04	62.92
40%	69.02	64.30 ← “expected”
30%	—	66.12
20%	—	68.34

⁶ This discussion is based on information in public sources about the Federal Trade Commission complaint issued July 27, 1977, against the *Los Angeles Times* quantity discount practices. Sources are: *Editor and Publisher*, “*L.A. Times* Reaches Accord with FTC in Ad Case,” November 8, 1980, “*L.A. Times* Pact Could Sharply Reduce Discounts,” November 15, 1980, “*L.A. Times* Seen Accepting FTC Way on Ad Discounts,” September 26, 1981, “*Los Angeles Times* to Fight Order,” October 3, 1981, “*L.A. Times* Explains Internal Effects of Ad Rate Agreement with F.T.C.,” October 10, 1981, “Why Volume Discounts are Justified,” January 9, 1982.

⁷ The example discussed here is given in Corey (1983) case study “International Business Machines (I).” The prices quoted are not public information and hence are probably disguised to some extent. However, the qualitative relationship of the prices mirrors the reality.

Thus, both firms offered discounts of approximately 10% in going from the bottom to top of the requirements level they were asked to bid on. The most common explanation of such discounts is that a supplier’s average unit cost declines with volume through amortization of fixed costs and/or learning curve effects. There is another possible explanation, however, which we develop here. We show that even with constant unit cost, quantity discounts may be optimal if the buyer has a desire to multisource procurement.

As in the IBM computer memory example above, we develop a somewhat simplified model of the situation. Consider a buyer wishing to purchase a total of three units of a good from two suppliers. The buyer asks firm 1 to provide prices for one and two units of supply; asks firm 2 to provide prices for one, two, and three units of supply. The buyer could conceivably “sole source” given these bids by purchasing all three units from firm 2. The buyer, however, may associate risks with sole sourcing e.g. supply disruption, power to the sole source supplier in future negotiations, etc. Consequently, there is a cost to sole sourcing in addition to the price paid to firm 2 for the three units. Let:

- S = buyer’s perceived cost of sole sourcing ($S \geq 0$),
- p_i = price quoted by firm 1 for i th unit, $i = 1, 2$,
- c_i = price quoted by firm 2 for i th unit, $i = 1, 2, 3$.

We take the perspective of firm 1 and ask under what conditions other than cost economies does optimality require $p_1 > p_2$, i.e., a quantity discount. To eliminate economy of scale effects, we assume firm 1 has constant unit cost which without loss of generality we take to be zero. Given price quotes from each firm, the buyer’s three alternatives and total cost associated with each are:

Units from		Total Cost
Supplier 1	Supplier 2	
0	3	$c_1 + c_2 + c_3 + S$
1	2	$p_1 + c_1 + c_2$
2	1	$p_1 + p_2 + c_1$

S appears in the first alternative only, since it is the only one with sole sourcing. Assuming the buyer’s objective is to minimize total cost, Table 2 summarizes the conditions on p ’s and c ’s, leading to each feasible choice. Table 2 arbitrarily assumes that ties are broken in firm 1’s favor. Figure 7 presents the results of Table 2 graphically by dividing the (p_1, p_2) plane into three mutually exclusive and exhaustive regions.

Figure 7 is similar to the plotting of a feasible region for a two-variable linear programming problem. $OABD$ is the region representing firm 1 selling two units. For this region, $p_1 + p_2$ is the correct objective function. The family of iso-profit lines for the $p_1 + p_2$ objective has the same slope as the line segment BD which makes a 45% angle with the x -axis as shown in Figure 7. Therefore, profits are maximized at all points on BD . Since B is the point on BD with maximum p_2 and minimum p_1 , if $p_1 > p_2$ at B , $p_1 > p_2$

TABLE 2
Feasible Options, Associated Firm 1 Profit and Buyer Decision Rules

Units Purchased Firm 1	Firm 1 Profit	Buyer Decision Rules
0	0	$p_1 > c_3 + S$ and $p_1 + p_2 > c_2 + c_3 + S$
1	p_1	$p_1 \leq c_3 + S$ and $p_2 > c_2$
2	$p_1 + p_2$	$p_2 \leq c_2$ and $p_1 + p_2 \leq c_2 + c_2 + S$

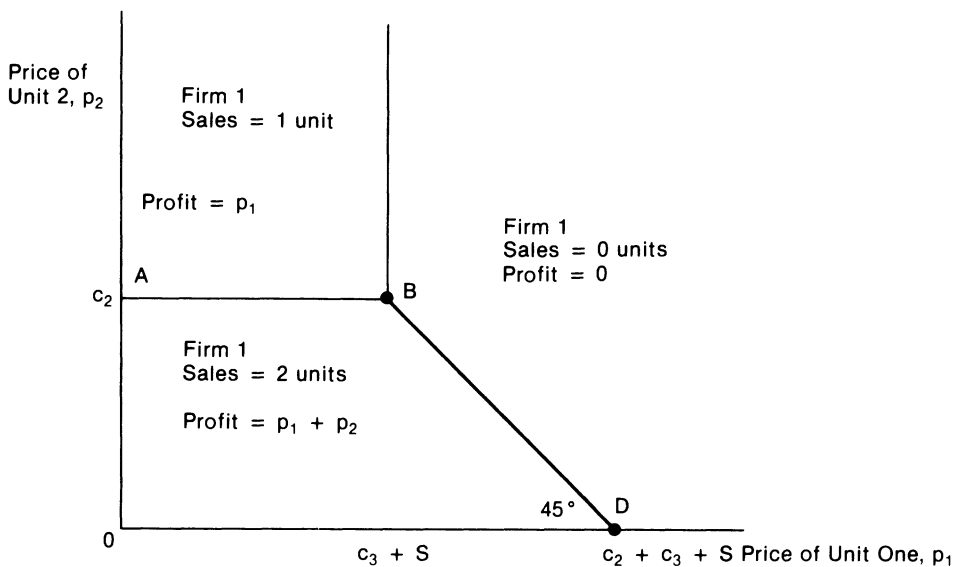


FIGURE 7. Firm 1 Sales as Function of (p_1, p_2) .

at all points on BD , i.e., optimality requires a quantity discount. The coordinates of point B in (p_1, p_2) space are $(c_3 + S, c_2)$. Therefore, quantity discounts are required if $c_3 + S > c_2$, i.e., S , the cost associated with sole sourcing is greater than $c_2 - c_3$.

As a special case, then, we see that if firm 2 has a uniform pricing policy ($c_1 = c_2 = c_3$), the optimal strategy for firm 1 involves quantity discounts if the buyer associates any strictly positive cost with sole sources.

Summary

These four cases demonstrate that quantity discounts can increase profits in several situations. The first three examples demonstrate quantity discount motivations established in the literature. The last example on competitive bidding provides a motivation previously not recognized in the literature. In addition to exemplifying the various motivations, the cases show key parameters of a discount schedule are:

- (i) Type, i.e., whether the schedule is incremental (Figure 1(c)) or all units (Figure 1(d));
- (ii) Complexity, i.e., the number of price breaks;
- (iii) Depth, i.e., the magnitude of the discounts;
- (iv) Qualifying unit base.

The range of profitable uses of quantity discounts shows that pricing managers must go beyond the simple “what price should I charge?” question. In addition, a regular part of pricing procedures of the firm should be the questions:

1. Should I offer quantity discounts?
2. If so, what should they be?
3. How can we administer a price schedule with quantity discounts?

The concluding section of the paper takes up these questions from the viewpoint of the manager.

5. From Theory to Practice

Many firms offer quantity discounts. Many firms do not. How can the pricing manager determine to which set his firm ought to belong? Because of its disregard of

implementation costs, the literature suggests that just about everybody should be offering quantity discounts, viz. a firm facing homogeneous buyers should have quantity discounts if:

- (i) buyers, either end users or resellers of the product, have downward sloping demand curves; or
 - (ii) inventory carrying and transportation costs are significant and buyers use some type of economic order quantity rule or
 - (iii) buyer wants to multisource procurement;
- those facing heterogeneous buyers should offer if:
- (iv) large quantity buyers are more price sensitive than small buyers or
 - (v) inventory/transportation cost situation is as described for homogeneous buyer case (ii).

The pricing manager must assess the extent to which these conditions hold in his situation. In general, the more pronounced each of these conditions is, the more benefit there is to be derived from a quantity discount structure. The extent of the inventorying/transportation cost motivation is typically easy to assess since the seller understands the logistics system well enough to assess users' costs. The price discrimination motivations, (i) and (iv), require the manager to assess the individual buyer's demand curve. In the heterogeneous buyer case, this theoretically means the firm must estimate not only the aggregate market demand curve (as it would to set the optimal uniform price), but also the individual level curves. In practice, the opportunity for partial price discrimination often reveals itself when a uniform price schedule is in effect. The buyer heterogeneity first becomes evident through very different purchase rates at the uniform price. Qualitative understanding of the individual customers is then used to assess if the necessary link between size and price sensitivity exists.

If the pricing manager finds sufficient justification for a quantity discount schedule, he must then ask if it would be possible to administer such a schedule. Quantity discounts require the firm to monitor and record purchases at the individual account level. In some cases, e.g., most consumer goods, this information burden is substantial and may preclude quantity discounts or, at best, restrict the firm to "quantity per order" based discounts. As discussed, a quantity per order discount is appropriate for the system efficiency motivation. However, price discrimination motivated discounts generally require aggregation over a time period and thus the ability to track purchases over time.

The second administration question is whether it is necessary to prevent resale from one user to another and, if so, is it possible to enforce this condition? The partial price discrimination motivation works only if the firm can structure a situation such that a large buyer paying a "low price" does not resell the good to a small buyer from whom the firm is trying to get a high price. The system efficiency motivation being based on cost savings is not subverted by resale.

Our two partial price discrimination examples, the *Los Angeles Times* and IBM, demonstrate two ways in which resale can be effectively prevented. First, the *Los Angeles Times* offers a service. Retail advertising space is not easily split up and resold, and the seller monitors who is consuming the service. More generally, if the product is perishable, its resale opportunities are limited. In the IBM case, after-the-sale service by the manufacturer is important. This creates a situation whereby the small firm wants to buy direct from the manufacturer even if a lower price is available from another source. A third way is for the firm to include "no resale" as part of its pricing terms and conditions. For example, a major footwear manufacturer has a "no transshipments" clause in its agreement with retailers stating "we have a policy of selling only directly to selected retail outlets for resale by them at such specified locations. Resale or transshipments of our merchandise to another location or to another business contravenes that

policy and the terms and conditions of sale and may result in nonshipment or termination of the retailer's business relationship with us." The firm must assess its ability to detect violations of this condition (several computer software manufacturers do this through coded warranty cards so the firm knows to whom it originally shipped the goods and the buyer fills in where the goods were purchased) and its willingness to terminate violators. If resale cannot be effectively prevented, a uniform price schedule may dominate despite the theoretical arguments for quantity discounts.

The issues of monitoring purchases at the individual level and preventing resale have been mentioned in the literature—usually as assumptions required to make the analysis hold. Two other practical issues of significance facing the pricing manager are: the timing of discounts and forward buying. The timing issue arises for all-units discounts with quantity aggregated over time, e.g. the *Los Angeles Times* situation above. A large buyer plans to qualify for the highest quantity/lowest price bracket. What price should he be charged for the first unit purchased? The buyer argues, with some justification, for the price he ultimately plans to qualify for. However, if the buyer does not qualify for this price by the end of the aggregation period, the firm must recoup the discount granted "up front." "Bill-back" procedures are more feasible in cases where the collection cost is small relative to the amount due and there is an ongoing relationship between the firm and the buyer, e.g. through required maintenance or repair. The firm obviously would prefer to charge high prices initially and grant rebates as required quantity levels are reached; however, large buyer resistance typically makes this infeasible.

The last issue which the manager must anticipate and manage is forward buying. Obviously, some forward buying is an intent of the system efficiency motivation. However, it can subvert the partial price discrimination motivation. If the good is easy to inventory, even "small" buyers will attempt to achieve the last quantity bracket by ordering several years' requirements at once. This excessive forward buying, unless anticipated and properly discouraged by the design of the schedule, can create a situation where the desired discrimination in quantity (and price) is not achieved. (Note that the *Los Angeles Times* and other service providers do not face this issue because the service cannot be inventoried.) Second, it can create problems in the firm's ability to supply each account's quantity demanded at a given point in time. Left unmanaged, a quantity discount increases the likelihood of stockouts. A stockout at the firm creates a problem in the price computation since buyers expect to be billed on the basis of quantity demanded, not quantity shipped. Thus, in an out-of-stock position, the firm faces the dilemma of defining "quantity" as quantity demanded rather than quantity supplied (thus losing the benefits sought in designing the schedule) or straining the account relation by not permitting the buyer access to the desired quantity bracket due to supply limitations.

Thus, before addressing the design issues, the pricing manager must weigh the potential benefits of a schedule vs. the possibility of effective implementation. The potential benefits are indicated by the extent of the conditions we noted indicating quantity discounts. Effective implementation depends upon: (i) monitoring sales at the account level, (ii) enforcing "no resale" condition if necessary for the motivation, (iii) resolving the timing of granting of discounts, and (iv) anticipating forward buying and balancing supply and demand at a given point in time. If the potential benefits are present and conditions make effective implementation possible, the pricing manager can use the guidelines from the literature to design a schedule. We now review the key design questions identified to state guidelines for managers and also to suggest future research areas. Table 3 summarizes the designs on schedule type, complexity, depth, and qualifying units for each firm in our study. We now review each design question in turn.

TABLE 3
Schedule Design Issue Resolution in Examples

	System Efficiency	Partial Price Discrimination		Bidding
	Sealed Air	IBM	L.A. Times	Suppliers
Type:	All Units	Incremental	All Units	—
Complexity:	6	4	33	5
Depth:	33%	12%	40%	10%
Qualifying Units:	Per Order	2-year Volume	Annual	Annual

Design Issue #1: Schedule Type

Our study gives examples of both all units (Sealed Air and *Los Angeles Times*) and incremental (IBM) schedules. (The components supplier case cannot really be classified as either because the buyer asked for five points on the total cost curve, $R(q)$, rather than specification of the entire curve.) The difference between incremental and all-units can be seen by examining the total cost curve for an all-units schedule as shown in Figure 8 for the all-units schedule p_1 if $q < x_1$, p_2 if $x_1 \leq q < x_2$, p_3 if $q \geq x_2$. The heavy line shows the effective price schedule. This effective schedule represents the fact that if a buyer purchased slightly less than a breakpoint quantity (x_1 or x_2), a firm typically charges the total cost associated with the breakpoint quantity if it is less.

The effective price schedule for an all-units type *must* have flat portions. An incremental schedule need not have a flat portion. Any all-units schedule can be written as an equivalent incremental schedule, but not vice versa. Leaving aside issues of schedule complexity and communication costs, the optimal incremental schedule is always as good as the optimal all-units schedule and sometimes better because it affords more flexibility in the total cost curve specification. This has been argued by Dolan (1978) for the system efficiency motivation and Oi (1971) for the partial price discrimination motivation. Oi shows that for a discount schedule with one-price break, the firm using an all-units type discount “fails to capture part of the consumer surplus that it could have appropriated by adopting a marginal price discount schedule [an incremental type schedule].” (p. 92)

Why then do Sealed Air and the *Los Angeles Times* employ all-units schedules? With respect to Sealed Air, the system efficiency literature examines continuous cost functions. Lal and Staelin (1984) use quadratic transportation costs but even with this advance we have yet to study the realities of the cost schedule facing Sealed Air, i.e., a sharply discontinuous one as transportation mode changes. As Figure 8 suggests, an all-units discount tends to “lump” orders at x_1 and x_2 which may be precisely the incentive Sealed Air wishes to give as x_1 and x_2 represent different boxcar sizes. The *Los Angeles Times*’ all-units specification is probably driven by communication costs. With 33 breaks in its price schedule, an incremental schedule would be difficult to compute and with so many breakpoints, the total cost function discontinuities are “small” anyway.

While the marketing and economics literature indicates the pricing manager should use incremental type schedules, this support stems from models in which the cost curves are continuous. It is an open research question whether optimality requires an all-units schedule with its flat portions for certain possibly discontinuous cost functions. While any all-units schedule can be rewritten as an incremental schedule, implementation costs argue for all-units schedules. Hence, if the optimal price schedule has “flats,” an all-units schedule may be preferred.

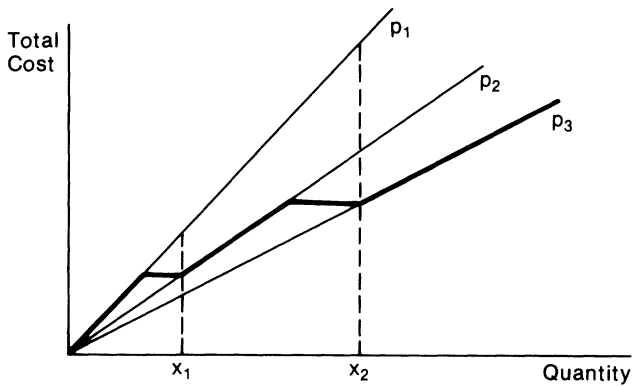


FIGURE 8. Total Cost for the All-Units Discount Schedule.

Design Issue #2: Schedule Complexity

The four examples vary widely in terms of schedule complexity, i.e., the number of price breaks in the schedule, ranging from 4 to 33. What determines the optimal number?

First, for both the system efficiency and price discrimination motivations, one finding is well established, viz. with homogeneous customers, a single price break is sufficient. Little has been done beyond this for the system efficiency case because the major work after Dolan’s formalization of the problem, Lal and Staelin (1984), takes $R(q)$ as a continuous function. Murphy (1977) provides the primary insight for the price discrimination case by deriving an expression for the marginal value of going from an “ N -break” to “ $N + 1$ -break” schedule. For the type of heterogeneity described above in Figure 5, he shows that the marginal value declines in $1/N^2$ where N is the number of price breaks. Hence, even though for this heterogeneity specification, the marginal value of adding a price break is always positive, the value declines sharply and there appears little added merit in 33 price-break schedules such as the *Los Angeles Times* over the more conventional four to six price-break models as employed by the other firms in the paper.

Recent work by Moorthy (1984) on product line pricing bears a structural similarity to the price discrimination work considered here. This work shows it is not just the degree of heterogeneity, but the specific type which is important. Specifically, while Murphy shows that $N + 1$ breaks are always strictly better than N for his heterogeneity specification, Moorthy shows that N customer segments differing in willingness to pay can, under some circumstances, be optimally served with $m < N$ products even if fixed costs are zero. Additional research is required, possibly using a model similar to Moorthy’s, to study the benefits of schedule complexity in various demand situations. Uncertainty in demand is a key factor to be studied. Managers frequently institute very simple schedules initially to see how buyers react. This added knowledge on price sensitivity is then used to structure more complex schedules. Research on what type of schedule yields maximal knowledge of buyers would be useful to impacting practice.

Design Issue #3: Schedule Depth

How large should the discounts be from one block to another? The range for our four firms was 10–40%. We know the answer in the homogeneous user case. For homogeneous users and the system efficiency motivation, Dolan (1980) and Monahan (1984) give closed-form expressions for the size of the discount in terms of the buyer’s demand rate and costs to order and hold inventory. For homogeneous buyers and the price discrimination motivation, Gabor (1955) shows the discounted price should be the

firm's marginal cost and the initial price set to extract all the consumer surplus. For the heterogeneous user case, the answer is less clear. Lal and Staelin (1984) examine the system efficiency motivation but provide only a numerical algorithm, no closed form solutions. For the price discrimination motivation, Faulhaber and Panzar (1977) and Murphy (1977) show the rate of decline depends on customer heterogeneity. As in the schedule complexity decision, however, our understanding is limited to specific types of heterogeneity.

Finally, for the competitive situation, only Oren, Smith and Wilson (1983) provide any insight. They show that as the number of competitors grows, the optimal price schedule becomes flatter, approaching price equal marginal cost for all quantities as the number of firms becomes infinitely large. In general, the use of quantity discounts in strategic competition is the area most needing study.

Design Issue #4: Qualifying Units

The firms in our examples used both quantity per order and quantity aggregated over a period of time. Quantity discounts based on the system efficiency motivation, such as Sealed Air, use quantity per order. This is necessary because the seller wishes to influence the timing of demand. The price discrimination motivation probably requires aggregation over some time period. Current practice seems to be an annual aggregation period, almost by default. The work reported here, however, points out the principle to be followed in selecting the aggregation period.

The basis for the price discrimination motivation is a negative correlation between "size" and willingness to pay. The qualifying units issue is essentially the question of how one defines "size." The answer is clear: define it in such a way that the desired negative correlation results. For example, consider buyer 1 needing three units every month and buyer 2 needing three units in January and July only. Our theory says quantity discounts are useful if buyer 1 is more price sensitive than buyer 2. If the aggregation period were defined as one month, buyers 1 and 2 would be the same "size"—buyer 2 just being "in the market" less often than buyer 1. Hence, a longer aggregation period is necessary to identify buyer 1 as the larger, more price-sensitive customer. For this design issue, then, the guidelines are clear and the specific situation dictates the answer.

In designing a price schedule, the pricing manager is usefully informed by the literature's establishing (i) the relationship between all-units and incremental schedules, (ii) the link between customer heterogeneity and schedule depth and complexity, and (iii) the principle to follow in setting the qualifying units. This section has also identified some key implementation issues which the manager must address in administering a quantity discount schedule.

Summary

This paper shows that pricing structures are important. The discounts in our four examples involve a significant portion of the firm's contribution dollars. Still, marketing managers have at best a vague understanding of quantity discounts. We wish to stress, however, that the scholarship on this subject both in economics and marketing is unambiguous on some points and relevant to managers.

Various issues of importance to marketing managers still remain unresolved, however. The broadest area for research is in strategic competition. The components supplier example shows discounts as a potential competitive weapon in bidding situations. We have yet to develop a normative model based on this motivation which gives an active role to both suppliers.

Second, we understand little about the cost side of quantity discounts, i.e., the cost of necessary information collection, administration, and communication. For example, do the benefits of complexity as seen in the *Los Angeles Times* case really outweigh the

costs or does it only seem that way to the firm because the costs are not easily measurable? These costs are a crucial part of the answer to the schedule type and complexity design questions.

Finally, we present all the major motivations for quantity discounts as we currently understand them. There is little doubt though that our list is not exhaustive. What are other possible motivations? Are there more discoveries of rationales for nonlinear price schedules still to be made? These research issues present the delightful duality of intellectual stimulation and managerial relevance.⁸

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⁸ This paper was received June 1984 and has been with the author for 4 revisions.

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